Chapter 4

HCF and LCM

4.1 Factors and Multiples

If a number x divides another number y exactly (without leaving any remainder), then x is a factor of y and y is a multiple of x.

Factors are set of numbers which exactly divides the given number.
Multiples are set of numbers which are exactly divisible by the given number.

For example: If the number is 8, then {8, 4, 2, 1} is the set of factors, while { 8, 16, 24, 32, ...} is the set of multiples of 8.

Note:

- Factors of a number are always less than or equal to the given number.
- Multiples of a number are always more than or equal to the given number.

4.2 Common Multiple

A common multiple of two or more numbers is a number which is completely divisible (without leaving remainder) by each of them.
For example: We can obtain common multiples of 3, 5 and 10 as follows:

- Multiples of 3 = {3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33 ...}
- Multiples of 5 = { 5, 10, 15, 20, 25, 30, 35, ...}

♦ Multiples of 10 = { 10, 20, 30, 40, 50, 60, ...}.

Common multiples of 3, 5 and 10 are { 30, 60, 90, 120 ...}

Least Common Multiple (LCM)

The LCM of two or more given numbers is the least number to be exactly divisible by each of them.
For example: We can obtain LCM of 4 and 12 as follows:

$$\text{Multiples of } 4 = 4, 8, \mathbf{12}, 16, 20, \mathbf{24}, 28, 32, \mathbf{36}, \ldots$$

$$\text{Multiples of } 12 = \mathbf{12}, \mathbf{24}, \mathbf{36}, \mathbf{48}, \mathbf{60}, \mathbf{72}, \ldots$$

Common multiples of 4 and $12 = 12, 24, 36, \ldots$

$\therefore$ LCM of 4 and 12 = 12

Methods to Calculate LCM

There are two methods to find the LCM of two or more numbers which are explained below.

Prime Factorisation Method

Following are the steps to obtain LCM through prime factorisation method.

Step I: Resolve the given numbers into their prime factors.

Step II: Find the product of all the prime factors (with highest powers) that occur in the given numbers.

Step III: This product of all the prime factors (with highest powers) is the required LCM.

Ex. 1 Find the LCM of 8,12 and 15.
Sol:

2 | 8
2 | 4
2 | 2
1

2 | 12
2 | 6
3 | 3
1

3 | 15
5 | 5
1

Factors of $8 = 2 \times 2 \times 2 = 2^3$.

Factors of $12 = 2 \times 2 \times 3 = 2^2 \times 3^1$.

Factors of $15 = 3 \times 5 = 3^1 \times 5^1$.

Here, the prime factors that occur in the given numbers are 2, 3 and 5 and their highest powers are 3 , 1 and 1 respectively.

$$\therefore \quad \text{Required LCM} = 2^3 \times 3^1 \times 5^1 = 8 \times 3 \times 5 = 120$$

Division Method

Following are the steps to obtain LCM through division method,

Step I Write down the given numbers in a line, separating them by commas.

Step II Divide them by a prime numbers which exactly divides atleast any two of the given numbers.

Step III Write down the quotients and the undivided numbers in a line below the 1st.

Step IV Repeat the process until you get a line of numbers which are prime to one another.

Step V The product of all divisors and the numbers in the last line will be the required LCM.

Ex. 2 What will be the LCM of 15, 24, 32 and 45?
Sol. LCM of 15, 24,32 and 45 is calculated as

2	15, 24, 32, 45
2	15, 12, 16, 45
2	15, 6, 8, 45
3	15, 3, 4, 45
5	5, 1, 4, 15
	1, 1, 4, 3

$$\therefore \quad \text{Required LCM} = 2 \times 2 \times 2 \times 3 \times 5 \times 4 \times 3 = 1440.$$

Note: Start division with the least prime number.

4.3 Common Factor

A common factor of two or more numbers is that particular number which divides each of them exactly.

For example: We can obtain common factors of 12, 48, 54 and 63 as follows:

Factors of 12 = $12, 6, 4, \mathbf{3}, 2, \mathbf{1}$

Factors of 48 = $48, 24, 16, 12, 8, 6, 4, \mathbf{3}, 2, \mathbf{1}$

Factors of 54 = $54, 27, 18, 9, 6, \mathbf{3}, 2, \mathbf{1}$

Factors of 63 = $63, 21, 9, 7, \mathbf{3}, \mathbf{1}$

$\therefore$ Common factors of 12, 48, 54 and 63 = $3, 1$

Highest Common Factor

HCF of two or more numbers is the greatest number which divides each of them exactly.
For example: 6 is the HCF of 12 and 18 as there is no number greater than 6 that divides both 12 and 18.
Similarly, 3 is the highest common factor of 6 and 9.

HCF is also known as **'Highest Common Divisor' (HCD)** and **'Greatest Common Measure(GCM)**.

Methods to Calculate HCF

There are two methods to calculate the HCF of two or more numbers which are explained below

Prime Factorisation Method

Following are the steps for calculating HCF through prime factorisation method,

Step I Resolve the given numbers into their prime factors.

Step II Find the product of all the prime factors (with least power) common to all the numbers.

Step III The product of common prime factors (with the least powers) gives HCF.

Ex. 3 Find the HCF of 24, 30 and 42.
Sol. Resolving 24,30 and 42 into their prime factors.

2	24
2	12
2	6
3	3
	1

2	30
3	15
5	5
	1

2	42
3	21
7	7
	1

Factors of $24 = 2 \times 2 \times 2 \times 3 = (2^3 \times 3^1)$.

Factors of $30 = 2 \times 3 \times 5 = (2^1 \times 3^1 \times 5^1)$.

Factors of $42 = 2 \times 3 \times 7 = (2^1 \times 3^1 \times 7^1)$.

The product of common prime factors with the least powers $= 2 \times 3 = 6$.

$\therefore$ HCF of 24, 30 and 42 = 6

Division Method

Following are the steps to obtain HCF through division method.

Step I Divide the larger number by the smaller one.

Step II Divide the divisor by the remainder.

Step III Repeat step II till the remainder becomes zero. The last divisor is the required HCF.

Note: To calculate the HCF of more than two numbers, calculate the HCF of first two numbers then take the third number and HCF of first two numbers and calculate their HCF and so on. The resulting HCF will be the required HCF of numbers.

Ex. 4 Find the HCF of 26 and 455.

Sol.

$$\begin{array}{r} 17 \\ 26\overline{)455} \\ \underline{26} \\ 195 \\ \underline{182} \\ 13 \end{array} \quad \Longrightarrow \quad \begin{array}{r} 2 \\ 13\overline{)26} \\ \underline{26} \\ 0 \end{array}$$

$\therefore$ Required HCF = 13

Ex. 5 What will be the HCF of 1785,1995 and 3381?

Sol. At the 1st step, use any two of three numbers.

$$\begin{array}{r} 1 \\ 1785\overline{)1995} \\ \underline{1785} \\ 210 \end{array} \quad \Longrightarrow \quad \begin{array}{r} 8 \\ 210\overline{)1785} \\ \underline{1680} \\ 105 \end{array} \quad \Longrightarrow \quad \begin{array}{r} 2 \\ 105\overline{)210} \\ \underline{210} \\ 0 \end{array}$$

$\therefore$ HCF for 1785 and 1995 = 105

At the 2nd step, use obtained HCF 105 and the 3rd given number 3381.

$$\begin{array}{r} 32 \\ 105\overline{)3381} \\ 315 \\ \hline 231 \\ 210 \\ \hline 21 \end{array} \quad \Longrightarrow \quad \begin{array}{r} 5 \\ 21\overline{)105} \\ 105 \\ \hline 0 \end{array}$$

$\therefore$ Required HCF = 21.

From the above examples we can notice that HCF is the factor of difference of the given numbers.

Note

To find the HCF of given numbers you can divide the numbers by their lowest possible difference. If these numbers are divisible by this difference, then this difference itself is the HCF of the given numbers otherwise any other factor of this difference will be its HCF.

Ex. 6 Find the HCF of 30, 42 and 135.
Sol. Notice that the difference between 30 and 42 is less than difference between 135 and 42 or 30.
Difference between 30 and 42 is 12, but 12 does not divide 30, 42 and 135 completely.
$\therefore$ Factors of 12 = 12, 6, 4, 3, 2, 1.
Clearly, 3 is the highest factor, which divides all the three numbers completely.
$\therefore$ 3 is the HCF of 30, 42 and 135.

4.4 Method to Calculate LCM and HCF of Fractions

The LCM and HCF can be obtained by the following formulae

$$\text{LCM of fractions} = \frac{\text{LCM of Numerators}}{\text{HCF of Denominators}}$$

$$\text{HCF of fractions} = \frac{\text{HCF of Numerators}}{\text{LCM of Denominators}}$$

Note:

1. All the fractions must be in their lowest terms. If they are not in their lowest terms, then conversion in the lowest form is required before finding the HCF or LCM.
2. The required HCF of two or more fractions is the highest fraction which exactly divides each of the fractions.
3. The required LCM of two or more fractions is the least fraction/integer which is exactly divisible by each of them.
4. The HCF of numbers of fractions is always a fraction but this is not true in case of LCM.

Ex. 7 Calculate the LCM of $\frac{72}{250}, \frac{126}{75}, \frac{162}{165}$

Sol. Here,

$$\frac{72}{250} = \frac{36}{124}, \frac{126}{75} = \frac{42}{25}, \frac{162}{165} = \frac{54}{55}$$

Thus, According to the formula,

$$\text{Required LCM } = \frac{\text{LCM of 36,42 and 54}}{\text{HCF of 125,25 and 55}} = \frac{756}{5}$$

Ex. 8 Find the HCF of $\frac{36}{51}, 3\frac{9}{17}$

Sol. Here

$$\frac{36}{51} = \frac{12}{17}, \; 3\frac{9}{17} = \frac{60}{17}$$

Thus, According to the formula,

$$\text{Required HCF of fractions } = \frac{\text{HCF of Numerators}}{\text{LCM of Denominators}} = \frac{\text{HCF of 12 and 60}}{\text{LCM of 17 and 17}} = \frac{12}{17}$$

Note: The LCM and HCF of decimals can also be obtained from the above formulae (by converting the decimal into fraction).

Ex. 9 Find the LCM of 0.6, 9.6 and 0.12.

Sol. The given numbers are equivalent to $\frac{6}{10}, \frac{96}{10}, \frac{12}{100}$

Now, we have to find the LCM of $\frac{6}{10}, \frac{96}{10}, \frac{12}{100}$

According to the formula,

$$\text{LCM of fractions} = \frac{\text{LCM of Numerators}}{\text{HCF of Denominators}}$$
$$= \frac{\text{LCM of } 6, 96, 12}{\text{HCF of } 10, 10, 100}$$
$$= \frac{2 \times 2 \times 3 \times 8}{10} = \frac{96}{10} = 0.96$$

2	6, 96, 12
2	3, 48, 6
3	3, 24, 3
8	1, 8, 1
	1, 1, 1

4.5 Fast Track Techniques to solve the questions

Technique 1

Product of two numbers = HCF of the numbers x LCM of the numbers

Ex. 10 The LCM of 2 numbers is 2079 and their HCF is 27. If the 1st number is 189, then find the 2nd number.

Sol. Here, LCM = 2079, HCF =27, 1st number = 189, 2nd number =?

According to the formula,

$$\text{Product of two numbers} = \text{HCF} \times \text{LCM}$$
$$\therefore \quad 189 \times \text{2nd number} = 27 \times 2079$$
$$\therefore \quad \text{2nd number} = \frac{27 \times 2079}{189} = 297$$

Technique 2

The greatest number which divides the numbers x, y and z, leaving remainders a, b and c respectively is given by *(This formula is true for any number of numbers)*

$$\text{HCF of } ((x-a), (y-b), (z-c)).$$

Ex. 11 Find the greatest number which divides 29,60 and 103 leaving remainders 5,12 and 7,respectively.
Sol. Given that,

$$x = 29 \qquad y = 60 \qquad z = 103$$
$$a = 5 \qquad b = 12 \qquad c = 7$$

Now, according to the formula,

$$\text{Required number } = \text{ HCF of } ((29-5),(60-12),(103-7)) = \text{ HCF of } 24, 48, 96$$

Now,

2	24
2	12
2	6
3	3
	1

2	48
2	24
2	12
2	6
3	3
	1

2	96
2	48
2	24
2	12
2	6
3	3
	1

$$\text{Factors of } 24 = 2\times 2\times 2\times 3 = 2^3\times 3^1$$
$$\text{Factors of } 48 = 2\times 2\times 2\times 2\times 3 = 2^4\times 3^1$$
$$\text{Factors of } 96 = 2\times 2\times 2\times 2\times 2\times 3 = 2^5\times 3^1.$$

$$\text{Required HCF of 24, 48 and 96 } = 2^3\times 3^1 = 8\times 3 = 24.$$

Technique 3

The least number which when divided by x, y and z leaves the remainders a, b and c respectively, is given by

$$\text{LCM of } (x, yz) - k \qquad \text{where} \quad k = (x-a) = (y-b) = (z-c)$$

[This formula is true for any number of numbers.]

Ex. 12 Find the least number which when divided by 24, 32 and 36 leaves the remainders 19, 27 and 31, respectively.
Sol. Given that,

$$x = 24 \qquad y = 32 \qquad z = 36$$
$$a = 19 \qquad b = 27 \qquad c = 31$$

Also

$$24-19=5, \quad 32-27=5, \quad 36-31=5$$

$$\therefore \qquad k=5$$

2	24, 32, 36
2	12, 16, 18
2	6, 8, 9
3	3, 4, 9
	1, 4, 9

According to the formula,

Required number = (LCM of 24, 32 and 36) − 5

Now, LCM of 24, 32 and 36 = $2\times2\times2\times3\times4\times3=288$

$\therefore$ Required number = 288 − 5 = 283

Technique 4

The least number which when divided by x, y and z leaves the same remainder k in each case, is given by

$$\text{LCM of } (x, y, z) + k$$

[This formula is true for any number of numbers.]

Ex. 13 Find the least number which when divided by 24, 30 and 54 leaves 5 as remainder in each case.

Sol. Given that, $x=24$, $y=30$, $z=54$ and $k=5$

According to the formula,

Required number = [LCM of (24, 30 and 54)] + 5

LCM of (24, 30 and 54) $=2\times3\times4\times5\times9=1080$

$\therefore$ Required number = 1080 + 5 = 1085

2	24, 30, 54
3	12, 15, 27
	4, 5, 9

Technique 5

The greatest number that will divide $x, y, z\ldots$ leaving the same remainder in each case, is given by

$$\text{HCF of } (|x-y|, |y-z|, |z-x|\ldots)$$

Ex.14 What is the greatest number that will divide 99, 123 and 183 leaving the same remainder in each case? Also, find the common remainder.

Sol. Given that, $x = 99$, $y = 123$ and $z = 183$

According to the formula,

$$\text{Required number } = \text{ HCF of } (|x-y|, |y-z|, |z-x|)$$

Now, $|x-y| = |99-123| = 24$

$|y-z| = |123-183| = 60$

$|z-x| = |183-99| = 84$

2	24
2	12
2	6
3	3
	1

2	60
2	30
3	15
5	5
	1

2	84
2	42
3	21
7	7
	1

Factors of $24 = 2^3 \times 3^1$

Factors of $50 = 2^2 \times 3^1 \times 5^1$

Factors of $84 = 2^2 \times 3^1 \times 7^1$

$\therefore$ HCF of 24,60 and $84 = 2^2 \times 3^1 = 12$

Common remainder

$$\frac{99}{12} = 8\frac{3}{12}, \qquad \frac{123}{12} = 10\frac{3}{12}, \qquad \frac{183}{12} = 15\frac{3}{12}$$

Clearly, the required common remainder= 3

Technique 6

When the HCF of each pair of n given numbers is a and their LCM is b, then the product of these numbers is given by

$$\text{HCF}^{n-1} \times \text{LCM} = a^{n-1} \times b$$

Ex. 15 There are five numbers. HCF of each possible pair is 4 and LCM of all the five numbers is 27720. What will be the product of all the five numbers?

Sol. Given, HCF $= 4$, LCM $= 27720$ and $n = 5$

According to the formula,

$\therefore$ Required product $= \text{HCF}^{n-1} \times \text{LCM} = 45^{5-1} \times 27720$

$= 4^4 \times 27720 = 256 \times 27720 = 7096320$

Useful Results

- If a is the HCF of two parts of b, then b must be divisible by a. This rule has been derived from the proven fact that, if two numbers are divisible by a certain number, then their sum is also divisible by that number.
- If the two numbers are prime to each other (coprimes), then their HCF should be equal to 1. Conversely, if their HCF is equal to 1, the numbers are prime to each other.
- LCM of numbers which are prime to each other is the product of numbers itself.
- If k is the HCF of p and q, then
  (i) k is the HCF of p and $p+q$
  (ii) k is the HCF of p and $q-p$
- LCM of numbers is always divisible by the HCF of given numbers.
  i.e., if x is the LCM of a, b, c, d, e and y is their HCF, then y divides x

Ex. 16 Is it possible to divide 1394 into 2 parts such that their HCF may be 34?

Sol. If 34 is the HCF of two parts of 1394, then 1394 must be divisible by 34.
Let us see

$$\frac{1394}{34} = 41$$

Hence, it is possible to divide 1394 into 2 parts such that their HCF may be 34.

Ex.17 Are 6 and 11 coprimes?

Sol. Clearly, HCF of 6 and 11 is equal to 1. Therefore, 6 and 11 are coprimes

or prime to each other.

$$6\overline{)11}\;\begin{array}{r}1\\ 6\\ \hline 5\end{array} \implies 5\overline{)6}\;\begin{array}{r}1\\ 5\\ \hline 1\end{array} \implies 1\overline{)5}\;\begin{array}{r}5\\ 5\\ \hline 0\end{array}$$

Technique 7

The greatest n-digit number which when divided by x, y and z

1. leaves no remainder.
   Required number = n-digit greatest number - R
2. leaves remainder k.
   Required number = (n-digit greatest number $-R$) + k

where, R is the remainder obtained when n-digit greatest number is divided by the LCM(x, y, z).

Ex. 18 Find the largest possible number of 5 digits which is exactly divisible by 32, 36 and 40.
Sol. Given, the numbers are 32,36 and 40.

$$1440\overline{)99999}\;\begin{array}{r}69\\ 8640\\ \hline 13599\\ 12960\\ \hline 639\end{array}$$

LCM of 32, 36 and 40 = 1440

Greatest 5-digit number = 99999

$\therefore$ Required number = 99999 − 639 = 99360

Technique 8

The smallest n-digit number which when divided by x, y and z leaves

1. no remainder.
   Required number = n-digit smallest number +($L-R$)

2. remainder k.
   Required number = n-digit smallest number $+(L-R)]+k$

where, R is the remainder obtained when n-digit smallest number is divided by LCM(x, y, z). L is the LCM(x, y, z).
Ex. 19 Find the least possible 5-digit number, which when divided by 10, 12, 16 and 18 leaves remainder 27.
Sol. First of all we will find the LCM of 10, 12, 16 and 18.

2	10, 12, 16, 18
2	5, 6, 8, 9
3	5, 3, 4, 9
	5, 1, 4, 3

LCM $= 2\times 2\times 3\times 5\times 4\times 3 = 720$

Smallest 5-digit number $= 10000$

On dividing 10000 by 720,

$$\begin{array}{r} 13 \\ 720\overline{)\,10000} \\ \underline{720} \\ 2800 \\ \underline{2160} \\ 640 \end{array}$$

$\therefore$ Required number $= 10000+(720-640)+27 = 10000+80+27 = 10107$

Method to Solve Questions Based on Bells

To solve such questions, following steps are used.

Step I Find the LCM of given time intervals.

Step II Obtained LCM is added to the initial time and result of this addition will be our answer (The next time when bells ring together).

Note: Before calculating LCM, make sure that all numbers are in the same unit. This method is also applicable to runner running round a circular track traffic light growing at different intervals.

Ex. 20 Seven bells ring at intervals of 2, 3, 4, 6, 8, 9 and 12 min, respectively.

They started ringing simultaneously at 5:00 in the morning. What will be the next time when they all ring simultaneously?
Sol. LCM of 2, 3, 4, 6, 8, 9 and 12

2	2, 3, 4, 6, 8, 9, 12
2	1, 3, 2, 3, 4, 9, 6
3	1, 3, 1, 3, 2, 9, 3
	1, 1, 1, 1, 2, 3, 1

$$\text{LCM} = 2 \times 2 \times 2 \times 3 \times 3 = 72 \text{ min} = 1 \text{ h } 12 \text{ min}$$

$\therefore$ Required time $= (5 + 1 : 12)$ O'clock $= 6 : 12$ O'clock in the morning.

Ex. 21 Six bells ring at intervals of 2,4,6,8,10 and 12 s, respectively. They started ringing simultaneously. How many times, will they ring together in 30 min?

$$\text{LCM}(2,4,6.8.10,12) = 120$$

i.e, Bells will ring together after every 120 $s = 2\ min$

$\therefore$ Required number of times $= \left(\frac{30}{2} + 1\right) = 16$